

TriadMarket

DAO-governed on-chain prediction market

Mukhametkali Dias | Qaldyqan Yerzat | Rudov Andrew

Blockchain Technologies 2 — Final Project

Scenario and product idea

Option D: binary outcome market with AMM pricing and oracle resolution

- Users trade YES/NO outcome shares for a future event.
- Outcome shares are ERC-1155 tokens created per market.
- Trading uses our own constant-product AMM with 0.3% fee.
- Resolution uses a Chainlink price feed through a staleness-checked adapter.

Core thesis

The project is not only smart contracts. It is a full protocol package: contracts, tests, audit, frontend, subgraph, CI, deployment scripts, and presentation.

Requirement coverage

Mapping the rubric to concrete implementation

- UUPS upgrade: UpgradeableTreasury V1 -> V2.
- CREATE + CREATE2: MarketFactory deploys deterministic markets and AMMs.
- Token standards: ERC20Votes/Permit, ERC-1155, ERC-4626.
- DeFi primitive: scratch-built CPMM with LP tokens.
- Oracle + governance + L2 deployment scripts included.

Architecture overview

The protocol is split into clear security domains

Tokens

- TRIAD governance token
- OutcomeToken ERC-1155
- LP token
- ERC-4626 fee vault

Protocol

- MarketFactory
- PredictionMarket
- OutcomeAMM
- ChainlinkPriceOracle

Governance

- Governor
- TimelockController
- Proposal lifecycle
- Treasury control

Critical authority moves to Timelock after deployment, so individual deployer/admin power is reduced.

Trading flow

How a trader uses the market

- Trader deposits collateral into PredictionMarket.
- Market mints equal YES and NO ERC-1155 outcome shares.
- Trader swaps one outcome for the other in OutcomeAMM.
- AMM enforces $x*y=k$ and min-output slippage protection.
- After resolution, winning shares redeem collateral.

Main invariant: after a swap, the reserve product should not decrease except for precisely accounted rounding and fee logic.

Governance flow

Full OpenZeppelin Governor stack

- TRIAD holders delegate voting power before proposing/voting.
- Voting delay: 1 day; voting period: 1 week.
- Quorum: 4%; proposal threshold: 1%.
- Timelock delay: 2 days before execution.
- Post-deployment script checks that no admin backdoor remains.

Security model

Defenses required by the rubric

- CEI and ReentrancyGuard are used on external state-changing flows.
- All ERC-20 transfers use SafeERC20.
- Privileged functions use Ownable or AccessControl.
- No tx.origin authorization; no transfer/send ETH pattern.
- Case studies reproduce and fix reentrancy and access-control vulnerabilities.

Oracle and resolution

Chainlink adapter protects market resolution

- Adapter checks `answer > 0`.
- Adapter checks `answeredInRound >= roundId`.
- Adapter rejects stale prices older than `maxAge`.
- Market resolves YES if price $\geq$ threshold after dispute window.
- This keeps oracle validation in one auditable place.

Frontend and subgraph

Full-stack protocol demonstration

- Wallet connection through MetaMask and WalletConnect.
- Reads TRIAD balance, voting power, delegate address, and market state.
- Supports state-changing actions: delegate, vote, deposit/swap flow.
- Subgraph indexes Market, Trade, LiquidityPosition, VaultDeposit, ProposalSnapshot.
- UI handles wrong network, rejected transactions, and insufficient balances.

Testing and CI

How the project is defended technically

- Unit tests target every public/external function and revert path.
- Fuzz tests cover swaps, vault actions, and voting power.
- Invariant tests cover AMM k, supply conservation, and treasury accounting.
- Fork tests are prepared for USDC, Uniswap router, and Chainlink feed checks.
- GitHub Actions runs build, tests, coverage, Slither, linting, and frontend build.

Team contribution

Each member owns a domain but must understand the whole system

- Dias: AMM, outcome token, vault, gas report.
- Qaldyqan: governance, oracle, upgradeability, audit report.
- Andrew: frontend, subgraph, CI/CD, deployment documentation.
- During Q&A, every member can explain the architecture and critical security assumptions.

Final defense message

What we want the instructor to remember

TriadMarket demonstrates the required Blockchain Technologies 2 domains in one coherent protocol: tokens, DeFi AMM, oracle safety, governance, upgradeability, indexing, frontend, DevOps, and security review.

The strongest defense angle: every major design choice is documented, tested, and connected to a rubric requirement.